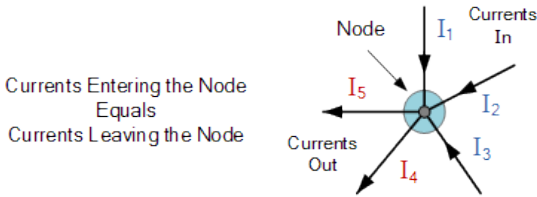
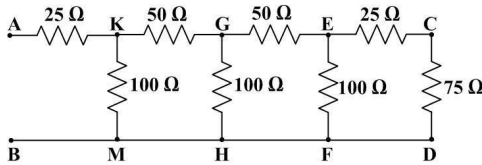
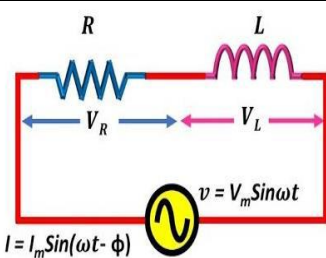
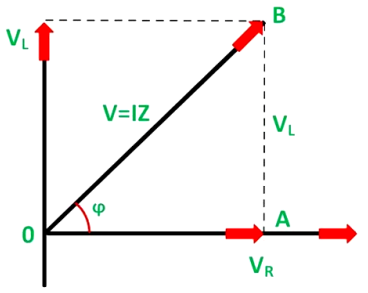
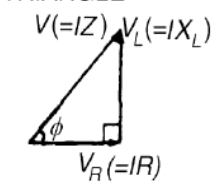
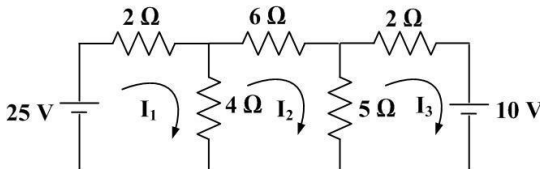
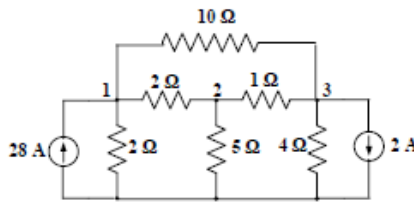
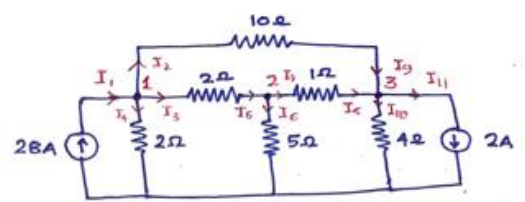
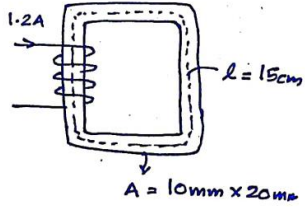


Sahrdaya College of Engineering and Technology, Kodakara		
Answer Key		
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY		
SECOND SEMESTER B.TECH DEGREE EXAMINATION, JUNE 2023		
Course Code: EST 130		
Course Name: BASICS OF ELECTRICAL AND ELECTRONICS ENGINEERING		
PART A		
		<i>Answer all questions. Each question carries 4 marks</i>
	Q1	State and explain Kirchhoff's laws.
	Ans	Kirchhoff's laws consist of two main principles: <i>Kirchhoff's Current Law (KCL):</i> KCL, also known as the junction rule, states that the algebraic sum of currents entering and leaving any node (or junction) in an electrical circuit is always zero.  $I_1 + I_2 + I_3 + (-I_4 - I_5) = 0$ KCL states that the total current entering a junction must equal the total current leaving the junction. This is a consequence of the conservation of electric charge, as charge cannot accumulate at a point in a closed circuit. <i>Kirchhoff's Voltage Law (KVL):</i> KVL, also known as the loop rule, states that the sum of the electromotive forces (EMFs or voltage sources) and the algebraic sum of the voltage drops (due to resistors and other circuit elements) in any closed loop in an electrical circuit is always zero. ie. $\sum IR + \sum \text{emf} = 0$
	Q2	Find the equivalent resistance across AB. 
	Ans	Resistance across EF = $100 // (25 + 75) = 50\Omega$ Resistance across GH = $100 // (50 + 50) = 50\Omega$ Resistance across KM = $100 // (50 + 50) = 50\Omega$ Resistance across AB = $25 + 50 = 75\Omega$

Q3	An alternating current is given by $I = 50\sin(314t)$. Find (a) the maximum value (b) frequency (c) time period of the current.
Ans	$I = 50\sin(314t)$ is of the form $I = I_m \sin(2\pi ft)$ Therefore a) Maximum value, $I_m = 50\text{A}$ b) $2\pi f = 314$ $f = 314/2\pi = 100/2 = 50\text{ Hz}$ Frequency, $f = 50\text{ Hz}$ c) $T = 1/f = 1/50 = 0.02\text{ s}$
Q4	Explain the phasor diagram and impedance triangle of a series resistive inductive circuit excited by an AC source.
Ans	  V : RMS value of applied voltage I : RMS value of resultant current V_R : Voltage drop across resistor $= IR$ V_L : Voltage drop across inductor $= IX_L = I\omega L$ VOLTAGE TRIANGLE  $V (=IZ)$ $V_L (=IX_L)$ $V_R (=IR)$ IMPEDANCE TRIANGLE  Z X_L R $V = \sqrt{V_R^2 + V_L^2}$ $= \sqrt{(IR)^2 + (IX_L)^2}$ $= I\sqrt{R^2 + X_L^2}$ $\sqrt{R^2 + X_L^2}$ is called Impedance (Z) of RL circuit $\phi = \tan^{-1}\left(\frac{V_L}{V_R}\right) = \tan^{-1}\left(\frac{X_L}{R}\right)$

Q5	Two impedances, $Z1 = (4+j3) \Omega$, $Z2 = (6 - j9) \Omega$ are connected in series. Find the equivalent impedance in polar form.
Ans	$Z1 = (4+j3) \Omega$, $Z2 = (6 - j9) \Omega$ Equivalent impedance = $(10 - j6) \Omega$ Equivalent impedance in polar form = $11.66 \angle -30.97$
PART B	
<i>Answer any one full question from each module. Each question carries 10 marks</i>	
Module 1	
Q6	Solve for the mesh currents in the given circuit. 
Ans	Consider Mesh 1 , By applying KVL $-2I_1 - 4(I_1 - I_2) + 25 = 0$ -----(1) In Mesh 2 , By applying KVL $-6I_2 - 5(I_2 - I_3) - 4(I_2 - I_1) = 0$ -----(2) In Mesh 3, Applying KVL $-2I_3 - 10 - 5(I_3 - I_2) = 0$ -----(3) Solving the above 3 equations, we get $I_1 = 4.891 \text{ A}$, $I_2 = 1.087 \text{ A}$, $I_3 = -0.652 \text{ A}$
Q7	Determine the node voltages in the given circuit. 
Ans	Considering Node 1 Applying KCL $I_1 = I_2 + I_3 + I_4$ $\text{i.e. } 28 = \frac{V_1 - V_2}{10} + \frac{V_1 - V_2}{2} + \frac{V_1}{2}$ $\Rightarrow V_1 - V_3 + 5V_1 - 5V_2 + 5V_1 = 280$ $11V_1 - 5V_2 - V_3 = 280 \longrightarrow \textcircled{1}$ 

		At Node 2, applying KCL $I_5 = I_6 + I_7$ $\frac{V_1 - V_2}{2} = \frac{V_2}{5} + \frac{V_2 - V_3}{1}$ $\Rightarrow 5V_1 - 5V_2 = 2V_2 + 10V_2 - 10V_3$ $\Rightarrow 5V_1 - 17V_2 + 10V_3 = 0 \rightarrow \textcircled{2}$ At Node 3, applying KCL $I_8 + I_9 = I_{10} + I_{11}$ $\frac{V_2 - V_3}{1} + \frac{V_1 - V_3}{10} = \frac{V_3}{4} + 2$ $20V_2 - 20V_3 + 2V_1 - 2V_3 = 5V_3 + 40$ $2V_1 + 20V_2 - 27V_3 = 40 \rightarrow \textcircled{3}$ Solving 3 equations, we get $V_1 = 36V, V_2 = 20V, V_3 = 16V$
Module 2		
Q8		a) A rectangular shaped core wound with a coil of 150 turns and 1.2A is made of mild steel plate 10 mm × 20 mm cross-section. The mean length of the magnetic path is 15cm. Calculate i. magnetizing force ii. flux density iii. reluctance iv. flux of magnetic circuit. Assume relative permeability of mild steel as 940. (6) b) State and explain Faraday's laws of electromagnetic induction. (4)
Ans	a)	$N = 150$ $I = 1.2A$ Area of cross section = $10\text{mm} \times 20\text{mm}$ $= 200\text{mm}^2$ $= 200 \times 10^{-6}\text{m}^2$ Length of magnetic path, $l = 15\text{cm}$ $= 15 \times 10^{-2}\text{m}$ Relative permeability of steel = 940 (μ_r) 

$$\mu_0 = 4\pi \times 10^{-7}$$

$$\begin{aligned} 1) \text{ Magnetising force, } H &= \frac{NI}{l} \\ &= \frac{150 \times 1.2}{15 \times 10^{-2}} = \underline{\underline{1200 \text{ AT/m}}} \end{aligned}$$

$$\begin{aligned} 2) \text{ Flux Density, } B &= \mu_0 \mu_r H \\ &= 4\pi \times 10^{-7} \times 960 \times 1200 \\ &= \underline{\underline{1.447 \text{ T}}} \end{aligned}$$

$$\begin{aligned} 3) \text{ Reluctance, } S &= \frac{l}{\mu_0 \mu_r A} \\ &= \frac{15 \times 10^{-2}}{4\pi \times 10^{-7} \times 960 \times 18 \times 200 \times 10^{-6}} \\ &= \underline{\underline{635248.67 \text{ AT/Wb}}} \end{aligned}$$

$$\begin{aligned} 4) \text{ Flux, } \phi &= B \times A \\ &= 1.447 \times 200 \times 10^{-6} \\ &= \underline{\underline{0.2894 \text{ mWb}}} \end{aligned}$$

- b) **Faraday's laws of electromagnetic induction** are fundamental principles underlying the operation of many electrical devices, including generators, transformers, and electric motors.. There are two main Faraday's laws

Faraday's First Law: The Change in Magnetic Flux Induces an EMF

Faraday's first law states that a change in magnetic flux through a closed loop of wire induces an electromotive force (EMF) or voltage in that loop. Magnetic flux (Φ) is the product of the magnetic field strength (B) and the area (A) through which the magnetic field lines pass, and it is defined as $\Phi = B \times A \times \cos(\theta)$, where θ is the angle between the magnetic field lines and the normal to the loop's surface. If the magnetic flux through the loop changes over time, it induces an EMF in the loop. This EMF causes an electric current to flow if the circuit is closed.

		Explanation: When a magnet is moved toward or away from a coil of wire or when the strength of the magnetic field passing through the coil changes, the magnetic flux through the coil changes. According to Faraday's first law, this change in magnetic flux induces an EMF in the coil. This induced EMF creates a potential difference across the ends of the coil, which can result in the flow of an electric current if the circuit is closed. <i>Faraday's Second Law:</i> The Magnitude of the Induced EMF is Proportional to the Rate of Change of Magnetic Flux Faraday's second law states that the magnitude of the induced EMF is directly proportional to the rate of change of magnetic flux through the loop. Mathematically, it can be expressed as: $E = -d\Phi/dt$ where: E is the induced EMF (voltage) in the loop. $d\Phi/dt$ represents the rate of change of magnetic flux with respect to time. Explanation: If the magnetic flux through a loop changes rapidly, the induced EMF will be greater, resulting in a stronger current. Conversely, if the magnetic flux changes more slowly, the induced EMF will be weaker, leading to a smaller current. The negative sign in the equation indicates that the induced EMF produces a current in a direction that opposes the change in magnetic flux, following Lenz's law.
	Q9	<i>Explain in detail the dynamically and statically induced emfs. An air solenoid has 300 turns, its length is 25 cm and cross sectional area of 3 cm². Calculate the self-inductance. If the coil current of 10A is completely interrupted in 0.04 s, calculate the induced emf in the coil.</i>
	Ans	a) EMF can be induced in 2 ways: 1) <i>Dynamically Induced EMF:</i> It is the EMF induced in a conductor or coil due to the relative motion between the conductor and the magnetic field. EMF will be induced dynamically when: ○ A conductor is moved across a stationary field ○ Field is moved across a stationary conductor

- Both the field and the conductor are moving

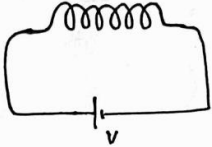
2) *Statically Induced EMF*: It is the EMF induced in a coil without any physical movement between the coil and the magnetic field.

Statically induced EMFs are of 2 types:

- Self-induced emf: It is the emf induced in a coil due to the change in current in the same coil itself is called self-induced emf.
- Mutually induced emf: It is the emf induced in a coil due to the change in current in the neighbouring coil is called mutually induced emf.

b)

No. of turns, $N = 300$
 Area of cross section, $A = 3\text{cm}^2$
 $\quad\quad\quad = 3 \times 10^{-4}\text{m}^2$
 Length, $l = 25\text{cm} = 25 \times 10^{-2}\text{m}$
 $\mu_0 = 4\pi \times 10^{-7}$
 Reluctance, $S = \frac{l}{\mu_0 A} = \frac{25 \times 10^{-2}}{4\pi \times 10^{-7} \times 3 \times 10^{-4}}$
 $\quad\quad\quad = \underline{\underline{663145596.2\text{ AT/Wb}}}$



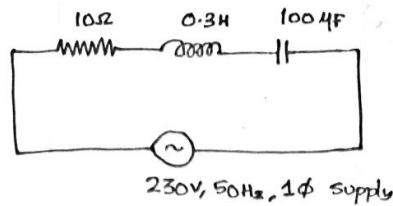
a) Self-Inductance, $L = \frac{N^2}{S} = \frac{[300]^2}{663145596.2}$
 $\quad\quad\quad = \underline{\underline{0.1357\text{ mH}}}$

b) Emf, $e = L \cdot \frac{di}{dt}$
 Current, $I = 10\text{A}$
 $t = 0.04\text{ sec}$
 $e = \frac{0.1357 \times 10^{-3} \times 10}{0.04} = \underline{\underline{0.034\text{ V}}}$

Module 3

Q10 A resistor of $10\ \Omega$, an inductor of $0.3\ \text{H}$ and a capacitor of $100\ \mu\text{F}$ are connected in series across a $230\ \text{V}$, $50\ \text{Hz}$, single phase ac supply. Determine (a) impedance (b) current (c) power in watts (d) circuit power factor.

Ans



$$R = 10\ \Omega$$

$$L = 0.3\ \text{H}$$

$$C = 100\ \mu\text{F} = 100 \times 10^{-6}\ \text{F}$$

$$\text{Resistance, } R = 10\ \Omega$$

$$\text{Inductive Reactance, } X_L = 2\pi fL$$

$$= 2 \times 3.14 \times 50 \times 0.3$$

$$= \underline{\underline{94.24\ \Omega}}$$

$$\text{Capacitive Reactance, } X_C = \frac{1}{2\pi fC} = \frac{1}{2 \times 3.14 \times 50 \times 100 \times 10^{-6}}$$

$$= \underline{\underline{31.84\ \Omega}}$$

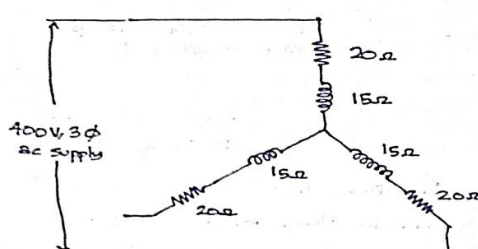
$$\begin{aligned} \text{(a) Impedance, } Z &= \sqrt{R^2 + (X_L - X_C)^2} \\ &= \sqrt{10^2 + (94.24 - 31.84)^2} \\ &= \underline{\underline{63.2\ \Omega}} \end{aligned}$$

$$\text{(b) Current, } I = \frac{V}{Z} = \frac{230}{63.2} = \underline{\underline{3.63\ \text{A}}}$$

$$\text{(c) Power, } P = VI \cos\phi$$

$$\text{(d) Power factor, } \cos\phi = \frac{R}{Z} = \frac{10}{63.2} = \underline{\underline{0.158\ \text{lag}}}$$

$$\begin{aligned} \therefore \text{Power, } P &= VI \cos\phi = 230 \times 3.63 \times 0.158 \\ &= \underline{\underline{132.27\ \text{W}}} \end{aligned}$$

Q11	Three impedances each having resistance 20Ω and an inductive reactance of 15Ω are connected in star across a $400V$, 3 phase, AC supply. Calculate (a) the line current (b) power factor (c) total power. If the load is connected in delta, determine the total power consumed by the load.
Ans	Given, $V_L = 400V$ $R = 20\Omega$ $X_L = 15\Omega$ Impedance/phase $Z_p = \sqrt{R^2 + X_L^2}$ $= \sqrt{20^2 + 15^2} = \underline{25\Omega}$  a) When connected in star, $V_L = \sqrt{3} V_{ph}$ $\text{or } V_{ph} = \frac{V_L}{\sqrt{3}} = \frac{400}{\sqrt{3}} = \underline{230.94 V}$ a) Line current = Phase current $= \frac{V_{ph}}{Z_p} = \frac{230.94}{25} = \underline{9.23 A}$ b) power factor, $\cos\phi = \frac{R}{Z} = \frac{20}{25} = \underline{0.8 \text{ lag}}$ c) Total power, $P = \sqrt{3} V_L I_L \cos\phi$ or $3 V_{ph} I_{ph} \cos\phi$ $= 3 \times 230.94 \times 9.23 \times 0.8$ $= \underline{5121.3 W}$ b) When connected in delta, $V_L = V_{ph} \therefore V_{ph} = 400V$ $I_{ph} = \frac{I_L}{\sqrt{3}} = \frac{400 (V_{ph})}{25} = \underline{16 A}$ $I_L = \sqrt{3} \times I_{ph} = \sqrt{3} \times 16 = \underline{27.71 A}$ $\cos\phi = R/Z = 0.8$ Total power = $3 \times V_{ph} \times I_{ph} \times \cos\phi$ $= 3 \times 400 \times 16 \times 0.8$ $= \underline{15360 W}$
